Supporting Information

Spontaneous Periodic Delamination of Thin Films to Form Crack-free Metal and Silicon Ribbons with High Stretchability

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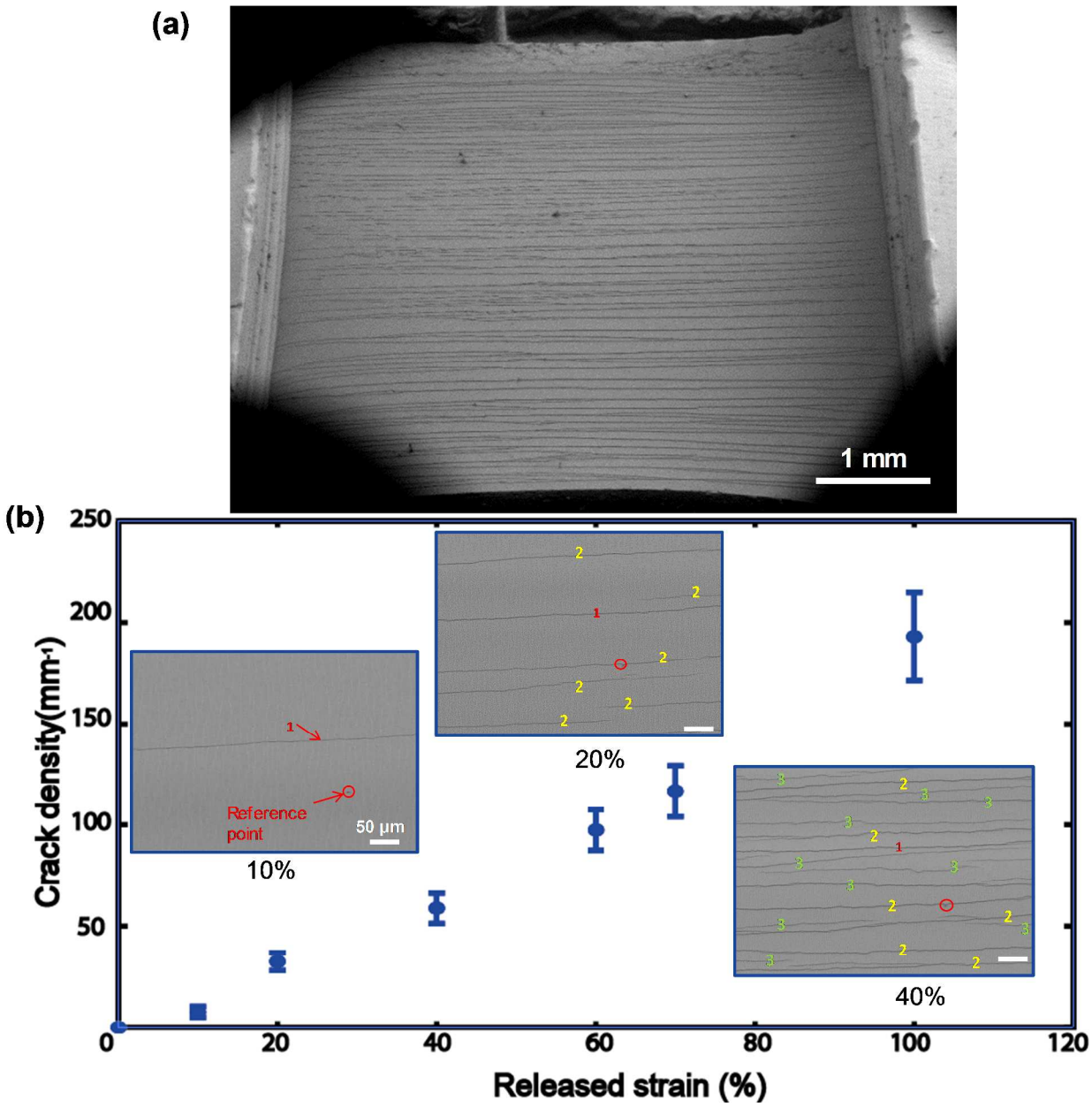


Figure S1: Highly ordered cracking fragmentation over the whole sample of buckled delaminated Au thin film on 100% pre-strained PDMS substrate. (a) The SEM image shows the cracking pattern in the whole Au film (t=40nm) of 5 mm × 4 mm after fully releasing the 100% pre-strain. (b) Experimental measurements of crack density (symbol) in Au film show that the crack density is linearly proportional to the percentage of the released pre-strain. Inset SEM

images show the increasing number of cracks at different level of released strain (10% (left), 20% (middle), 40% (right), where 1, 2, 3 represent the sequence of the occurrence of cracks.

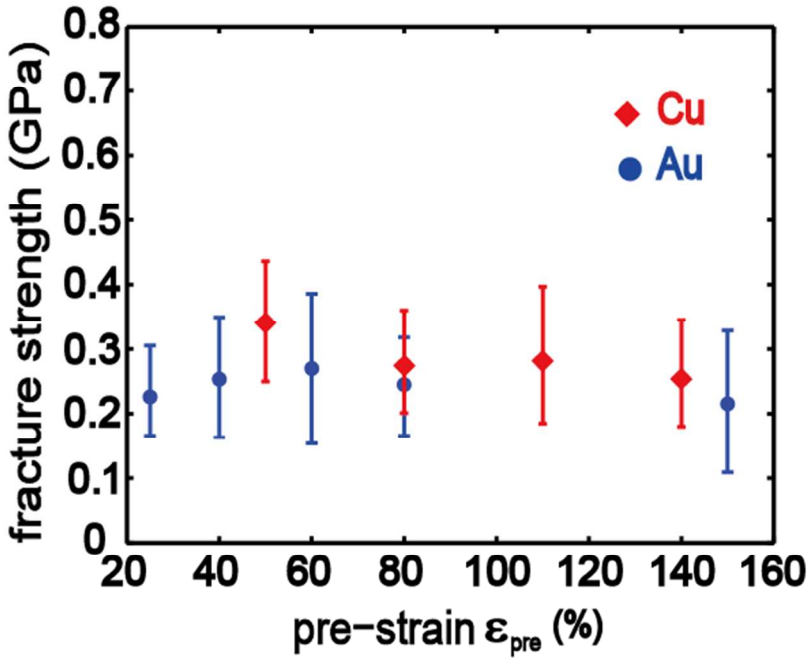


Figure S2: Estimation of the fracture strength for Au and Cu thin film from the measured cracked fragment width. From Equation S2, σ_c can be estimated as $d\ E_s(\upsilon\varepsilon_{pre}-\varepsilon_c)/2t$ with the measurement of cracked fragment width. The plot of estimated strength vs. different levels of pre-strain shows an almost constant value with a slight perturbation.

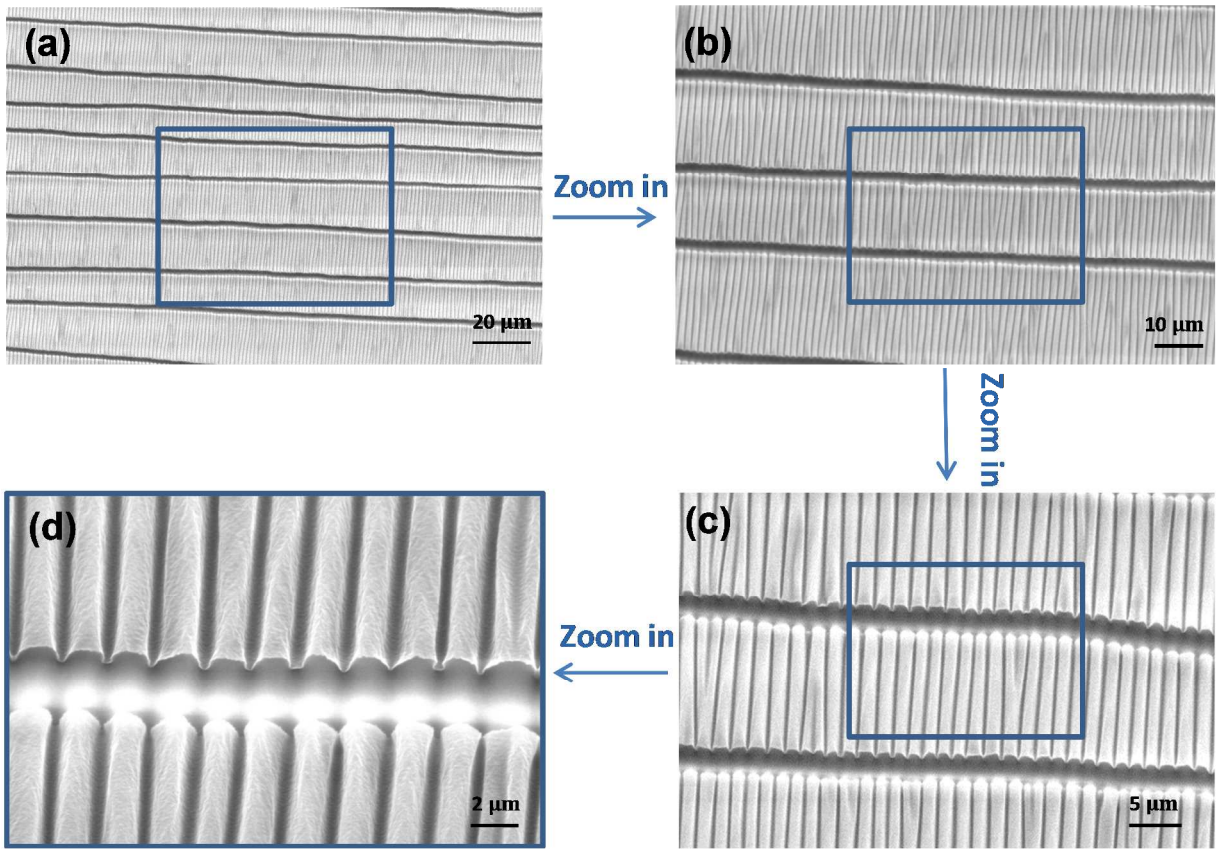


Figure S3: No longitudinal cracks along the delaminated blisters observed in Au film (25nm thick) on 150% pre-strained PDMS substrate upon pre-strain full release. (a) SEM image of the cracking pattern generated in a 200µm × 120µm large area. The blue rectangular area is zoomed in for (b) 100µm × 60µm and (c) 50µm × 30µm. (d) The largest magnification of SEM image up to 2µm scale shows the profile of spontaneously buckling-driven delamination and no longitudinal cracks observed perpendicular to the horizontal stretching direction.

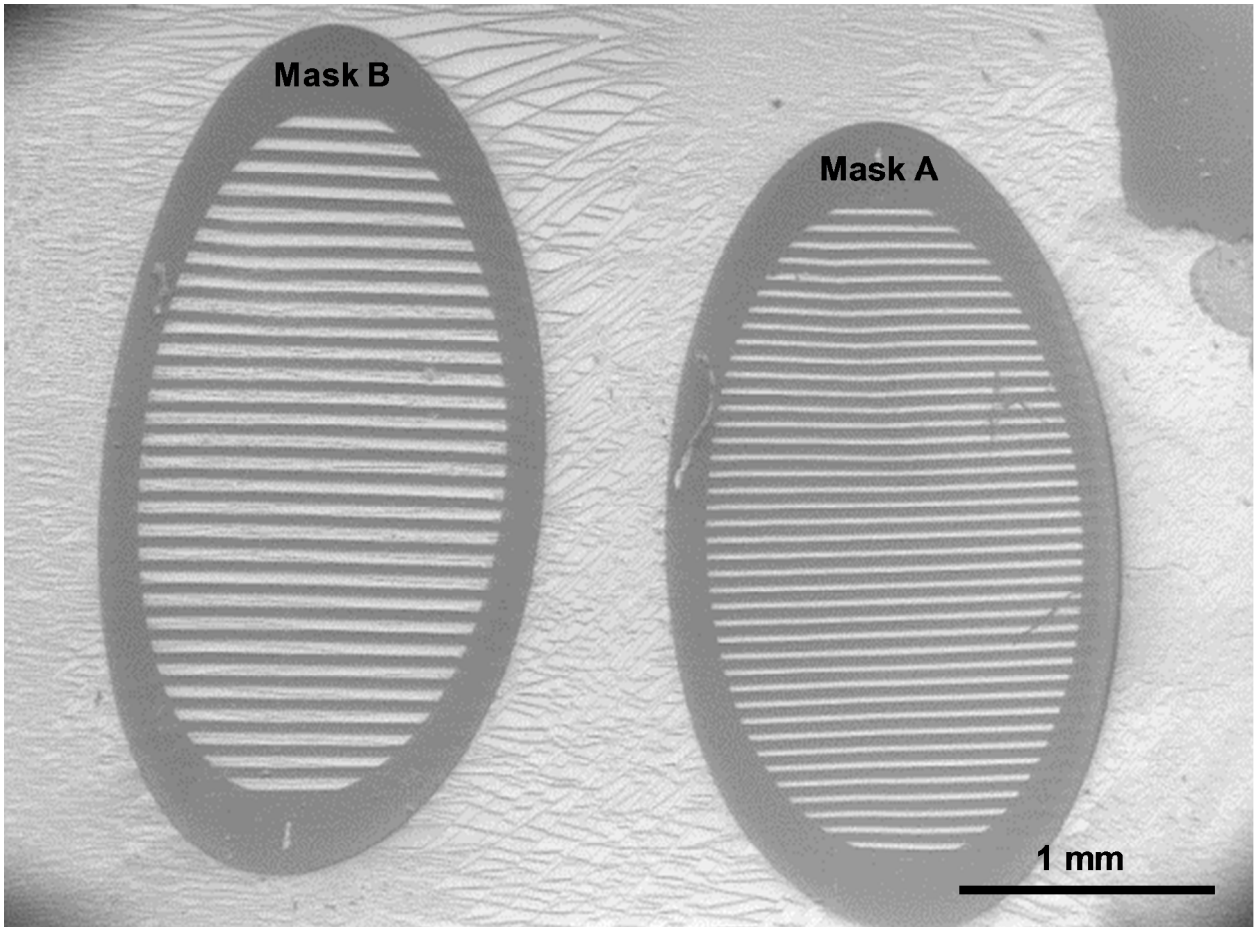


Figure S4: The cracking patterns over the whole sample of two different types of Au ribbons (40nm thick) fabricated with two shadow masks on a 100% pre-strained PDMS substrate. The ribbon width for Mask A and B is 22μm and 48μm, respectively. For ribbons with Mask A, no cracks are observed in all ribbons. For ribbons with Mask B, one crack is observed in most of the ribbons. Outside the masks, disordered cracking fragmentations are observed.

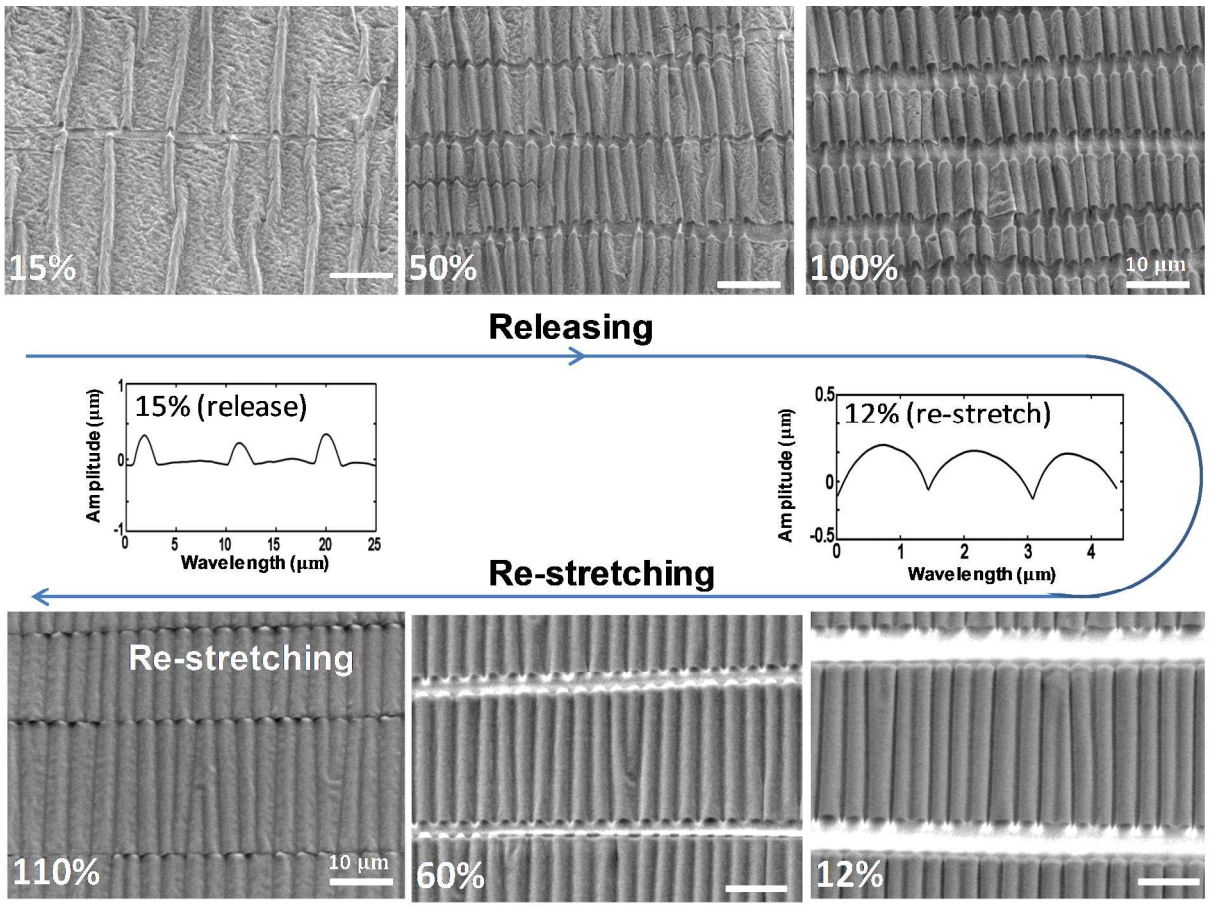


Figure S5: One reversible cycle of mechanical strain releasing and re-stretching in the spontaneously buckled delaminated Si film on 120% pre-strained PDMS substrate. Top row: the SEM images of the evolution of delaminated and cracked buckled patterns with the released strain from 15% to 100%. Bottom row: Evolution of the patterns after re-stretching the fully released sample from 12% to 110%. The middle row show the AFM measured profiles of the delaminated buckles at a released strain of 15% (left), and a re-stretching strain of 12% (right).

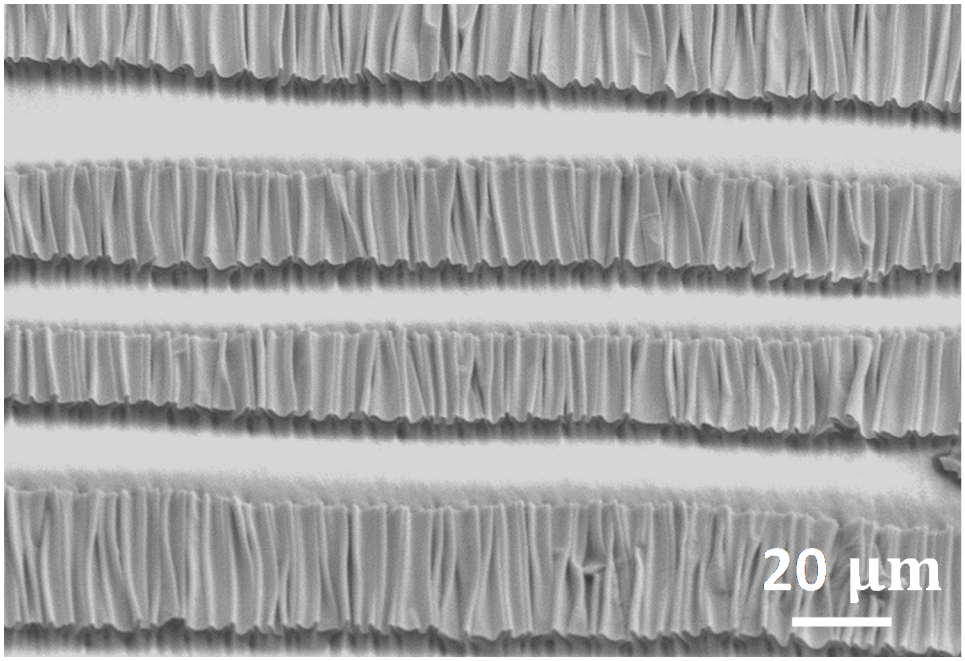


Figure S6: No longitudinal cracks along the delaminated blisters observed in Au film (40nm thick) on 400% pre-strained silicone rubber substrate upon pre-strain full release. SEM image of spontaneously buckled delaminated Au thin film on 400% pre-strained silicon rubber substrate.

Crack space in the buckling driven delamination patterns of thin films

When applying tension to a two-layer composite, a rubber substrate covered by a rigid platinum film, Volynskii et al.,[S1] showed the multiple cracking on the platinum film. By assuming a maximum tensile stress develops at the midpoint between cracks, Volynskii et al., gives the average width of fragments d as below[S1]:

$$d = \frac{2t\sigma_c}{E_s(\varepsilon-\varepsilon_c)} \quad \text{(S1)}$$

where t and σ_c are the thickness and fracture strength of the thin film, respectively. E_s is the Young's modulus of the substrate. ε and ε_c are the applied tensile strain and the onset fracture strain of the film, respectively.

During the release of substrate pre-strain in our studied system, we observed that subsequent cracks approximately divide the fragments at their midpoints, which is consistent with the assumption that the developed maximum tensile stress is located at the midpoint of the fragment. Similarly, by replacing the applied tensile strain in Eq. (S1) with the tensile strain due to the Poisson's effect in our pre-strained substrate system, the average width of fragment can be modified as

$$d = \frac{2t\sigma_c}{E_s(\upsilon\varepsilon_{pre} - \varepsilon_c)} \tag{S2}$$

It should be noted that Eq. (S1) assumes that the film remains well bonded to the substrate. In our studied system, despite the delamination of the film from the substrate, we observe that there is little slip at the film-substrate interface along the direction of delaminated blisters, which could be negligible. Thus, Eq. (S2) could still be used to predict the crack space in the buckling driven periodic delamination patterns of thin films on a pre-strained substrate.

Geometry of periodic buckled delaminated blisters:

The deflection of thin film can be approximated by a sinusoidal profile as

$$y = \frac{A}{2}\left(1 + cos\frac{2\pi}{\lambda}x\right) \tag{S3}$$

where λ and A are the wavelength and amplitude of blisters, respectively. For delaminated buckles with strip width w_f, the total potential energy of the film and substrate system can be expressed as:

$$U_{total} = w_f[\frac{1}{2}\int_0^{\lambda}\frac{\overline{E_f}t^3}{12}\left(\frac{d^2y}{dx^2}\right)^2 dx + \frac{1}{2}\int_0^{\lambda}\overline{E_s}\varepsilon^2\lambda dx + \Gamma\lambda] \quad \text{(S4)}$$

Here Γ is interfacial toughness. ε is the applied strain with $\varepsilon = \varepsilon_{pre}/(1+\varepsilon_{pre})$. The total potential energy is the summation of the bending energy in the thin film, the strain energy stored in the substrate, as well as the adhesion energy between the film and substrate. By minimization of the total potential energy with amplitude A and wavelength λ, the size of buckle-delamination blisters can be obtained.

References

S1 a) A. L. Volynskii, S. Bazhenov, O. V. Lebedeva, A. N. Ozerin, N. F. Bakeev, *Journal of Applied Polymer Science* **1999**, 72, 1267; b) S. L. Bazhenov, A. L. Volynskii, V. M. Alexandrov, N. F. Bakeev, *Journal of Polymer Science Part B: Polymer Physics* **2002**, 40, 10.